CERTIFIED $x^{0.90}$ LOWER BOUNDS FOR COLLATZ PREDECESSOR SETS

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Abstract. Let

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ (3n+1)/2, & n \equiv 1 \pmod 2, \end{cases}$$

and let $\pi_a(x)$ count the positive integers $n \le x$ whose T-orbit reaches a fixed target a. Krasikov and Lagarias proved in 2003 that $\pi_a(x) \ge x^{0.84}$ for every positive a not divisible by 3, once x is sufficiently large. A preceding level-15 exact certificate and formalization raised the displayed exponent to 0.88. Here we prove

$$\pi_a(x) \ge x^{9/10} = x^{0.90}$$

under the same hypotheses. More precisely, for every such a there is a constant $C_a > 0$ for which $\pi_a(x) \ge C_a x^{901/1000}$ eventually.

The proof combines an exact feasible point for the level-18 Krasikov–Lagarias linear program with a new adaptive advanced-term elimination. The successful policy retains a least-potential lift when all ordinary auxiliary lifts are dominated; this replaces the globally fixed fallback used at level 15. Two finite objects are checked: a vector satisfying 129,140,163 linear-program rows and a bounded adaptive potential satisfying 215,233,605 transition inequalities. All 344,373,768 row and transition inequalities are checked with exact integer arithmetic. The complete theorem is formalized in Lean. Its two explicit native-computation steps are also replayed in full by separate exact Python and C++ verifiers.

Generative-AI disclosure. Generative AI played a substantial role in the discovery and derivation of the mathematics in this paper, not merely in editing its presentation. Lech Mazur served as curator and prompter, designed and authored the orchestration and verification harness in which the models were run, selected and reconciled their outputs, and takes responsibility for the resulting manuscript and its claims. The agentic environments used included OpenAI Codex and Anthropic Claude Code; the models used included GPT-5.5 and Fable 5. These systems contributed to mathematical exploration, proof development, formalization, computation, and exposition. The principal theorem is supported by the Lean formalization and exact certificate checks described in the paper, and responsibility for any remaining errors lies with the human author.

Theorem and formal proof on Proof Atlas

1. Introduction

The Collatz problem is usually phrased as a question about the forward orbit of one integer. A complementary viewpoint fixes a target a and studies the size of its predecessor set. For the one-division map T in the abstract, define

$$\pi_a(x) = \#\left\{ n \in \mathbb{N} : 1 \le n \le x,\ T^j(n) = a \text{ for some } j \ge 0 \right\}. \tag{1}$$

This measures how much of the positive integer tree has merged into the orbit of a below height x. Such lower bounds do not prove the Collatz conjecture, but they give unconditional quantitative information about its inverse dynamics.

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The condition $3 \nmid a$ is forced by the problem, not by our method. If $3 \mid a$, an odd number cannot map to a. Iterating the unique even inverse therefore shows that the predecessors are exactly $\{2^r a : r \geq 0\}$, and consequently $\pi_a(x) = \Theta(\log x)$.

The published exponent chain is

$$.05 \longrightarrow .30 \longrightarrow 3/7 \longrightarrow .48 \longrightarrow .65 \longrightarrow .81 \longrightarrow .84.$$

The first two values are due to Crandall and Sander [3, 10]; Krasikov introduced the difference-inequality method [4], and Wirsching obtained the 0.48 bound [12]. Applegate and Lagarias proved the 0.65 tree-search bound and the 0.81 Krasikov-inequality bound [1, 2]. Krasikov and Lagarias then obtained exponent 0.84 from the no-truncation program at level 11 [5, Theorem 6.1 and Table 2]. Broader background is given in [6, 13].

Our main result is the following.

Theorem 1.1 (Main theorem)**.** *Let a be a positive integer with $3 \nmid a$. Then there exists $x_0 = x_0(a)$ such that*

$$\pi_a(x) \geq x^{9/10}$$

for every real $x \geq x_0$.

The proof first gives a slightly stronger exponent with a target-dependent constant.

Theorem 1.2 (Constant-factor form)**.** *For every positive integer a with $3 \nmid a$, there are constants $C_a > 0$ and X_a such that*

$$\pi_a(x) \geq C_a x^{901/1000}$$

for every $x \geq X_a$.

Since $901/1000 - 9/10 = 1/1000$, the main theorem follows once $C_a x^{1/1000} \geq 1$.

The theorem strengthens the author's preceding checked level-15 result $\pi_a(x) \geq x^{0.88}$. The present finite state space is exactly $3^3 = 27$ times larger, but the advance is not merely a larger computation. The level-15 termination proof used lift zero whenever all three auxiliary lifts had been marked dominated. At level 18 the corresponding fixed-policy work-list does not yield the bounded potential required by that proof. The new argument selects a fallback lift from the state-dependent potential itself and proves that this adaptive policy is compatible with both coefficient transport and the time-local functional minimum.

The proof has five interfaces.

(1) Inverse branches give the difference system equations (5) to (7) for predecessor envelopes.
(2) An exact level-18 vector satisfies the coefficient linear program at $\lambda = 2^{901/1000}$.
(3) A history-sensitive expansion and an adaptive finite potential produce finite normal forms whose terminal shifts are uniformly negative.
(4) At each residue and time, a critical pruning follows lifts that actually attain the functional minima. This preserves the functional inequality and can only improve the coefficient inequality.
(5) A choice-valued retarded induction gives exponential envelope growth; target reduction and the exponent reserve then yield theorem 1.1.

The proof as presented is not numerically effective. It proves that finitely many normal forms exist, and hence that common retarded shift bounds exist, without computing their maximal depth or an explicit $x_0(a)$. An effective extraction is possible in principle at fixed level, but no numerical threshold is claimed here.

Computer assistance. There are two large finite lemmas. One certifies the level-18 feasible point; the other certifies the adaptive termination potential. Their elementary integer predicates are stated in sections 3 and 4.4. Everything after these two inputs is deductive. The entire chain is formalized in Lean, as detailed in section 7.

Attribution. The envelopes, difference inequalities, and linear-program framework come from [5]. We supply a self-contained elimination argument with explicit history, adaptive branch selection, uniform retardation, and time-local functional interfaces. This choice is methodological; the paper makes no claim about the correctness, completeness, or revision history of any other proof.
Priority status. A dated search through July 15, 2026 found no published improvement of the 0.84 all-target exponent after 2003; a December 2025 preprint also describes 0.84 as the historical record [7]. The preceding 0.88 release and the present 0.90 theorem are therefore, to the best of our knowledge, the first improvements in this family since 2003. This is subject to the possibility of an overlooked publication or private computation.

2. Predecessor envelopes and difference inequalities

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, while orbit sources and targets are positive unless stated otherwise. Write T^j for the j-fold iterate. We retain the bounded-orbit normalization used in the Krasikov–Lagarias framework because it makes all inverse branches live under one common cutoff.

2.1. Bounded predecessors.

Define the bounded-orbit count

$$\pi_a^*(x) = \#\left\{n \in \mathbb{N} : 1 \le n \le x,\ T^j(n) = a \text{ for some } j,\ T^i(n) \le x\ (0 \le i \le j)\right\}. \tag{2}$$

Evidently $\pi_a^*(x) \le \pi_a(x)$. Call a *periodic* if $T^r(a) = a$ for some $r \ge 1$.

Fix $k \ge 2$. For a residue $m \pmod{3^k}$, let $\mathcal{A}_k(m)$ be the set of positive, nonperiodic integers congruent to $m \pmod{3^k}$. When this set is nonempty, put

$$\phi_k^m(y) = \inf_{a \in \mathcal{A}_k(m)} \pi_a^*(2^y a), \qquad y \ge 0. \tag{3}$$

The values are positive integers, so every nonempty infimum is attained. We use the principal residue set

$$\mathcal{P}_k = \left\{m \pmod{3^k} : m \equiv 2 \pmod 3\right\}, \qquad |\mathcal{P}_k| = 3^{k-1}.$$

Lemma 2.1 (Envelope properties)**.** *For $m \in \mathcal{P}_k$:*

(P1) $\phi_k^m(y) \ge 1$ for every $y \ge 0$;
(P2) ϕ_k^m is nondecreasing;
(P3) if $r \equiv 2 \pmod 3$ is a residue modulo 3^{k-1}, then

$$\phi_{k-1}^r(y) = \min_{0 \le \ell < 3} \phi_k^{r+\ell 3^{k-1}}(y). \tag{4}$$

Proof. The target itself is counted in $\pi_a^*(2^y a)$, giving (P1). Increasing the cutoff can only add bounded predecessors, giving (P2). Finally, the targets in one residue class modulo 3^{k-1} partition into the three displayed classes modulo 3^k. The infimum over that union is the minimum of the three attained integer infima, which gives (P3). □

At level 18 all principal target classes are nonempty. Indeed, 2 has order $2 \cdot 3^{17}$ modulo 3^{18}, so every unit class contains a power 2^e. Taking $e \ge 2$, such a power is nonperiodic because its orbit reaches the cycle $1 \leftrightarrow 2$ without returning to 2^e.

2.2. The three difference inequalities.

Put

$$\alpha = \log_2 3.$$

Residues below are reduced in their indicated moduli. The inverse branches of T, together with the bounded-orbit restriction, give Krasikov's difference system.

Proposition 2.2 (Difference system). *Let $k \geq 2$, $m \in \mathcal{P}_k$, and $y \geq 2$. Then*

$$m \equiv 2 \pmod 9 : \quad \phi_k^m(y) \geq \phi_k^{4m}(y-2) + \phi_{k-1}^{(4m-2)/3}(y+\alpha-2), \tag{5}$$

$$m \equiv 5 \pmod 9 : \quad \phi_k^m(y) \geq \phi_k^{4m}(y-2), \tag{6}$$

$$m \equiv 8 \pmod 9 : \quad \phi_k^m(y) \geq \phi_k^{4m}(y-2) + \phi_{k-1}^{(2m-1)/3}(y+\alpha-1). \tag{7}$$

The lower-level terms are interpreted by equation (4).

Proof. We spell out the counting and cutoff normalization. For a nonperiodic target $a \equiv 2 \pmod 3$, one inverse branch is $4a$, and

$$2^{y-2}(4a) = 2^y a.$$

The odd inverse is $(2a-1)/3$. In a D3 row it belongs to the required lower-level class and

$$2^{y+\alpha-1}\frac{2a-1}{3} \leq 2^y a, \qquad 2^\alpha = 3.$$

In a D1 row, insert one extra even step; the source is $2(2a-1)/3$ and the exponent is $y+\alpha-2$. These inverse branches remain inside the common cutoff and map to a. Their predecessor families are disjoint when a is nonperiodic. Indeed, a common source would visit both branch roots in some chronological order. The first root reaches a in at most two steps; continuing to the second root and then back to a would therefore give a positive return time for a. The branch targets are nonperiodic by the same deterministic-orbit argument. Taking the minimum over targets gives D1 and D3; discarding the odd branch gives D2. □

The signs of the shifts expose the analytic difficulty: $\alpha - 2 < 0$, whereas $\alpha - 1 > 0$. Thus a D3 auxiliary term asks for an envelope value at a later time. A direct induction is impossible until this advanced argument has been expanded into a finite expression whose terminal shifts are all negative.

3. The level-18 linear program and exact certificate

Fix $k \geq 2$, $1 < \lambda < 2$, and positive principal weights c_m, $m \in \mathcal{P}_k$. For a lower-level residue r, define

$$\underline{c}_r = \min_{0 \leq \ell < 3} c_{r+\ell 3^{k-1}}. \tag{8}$$

The no-truncation program $L_k^{NT}(\lambda)$ consists of

$$m \equiv 2 \pmod 9 : \quad c_m \leq \lambda^{-2} c_{4m} + \lambda^{\alpha-2}\underline{c}_{(4m-2)/3}, \tag{9}$$

$$m \equiv 5 \pmod 9 : \quad c_m \leq \lambda^{-2} c_{4m}, \tag{10}$$

$$m \equiv 8 \pmod 9 : \quad c_m \leq \lambda^{-2} c_{4m} + \lambda^{\alpha-1}\underline{c}_{(2m-1)/3}. \tag{11}$$

The original linear-program formulation treats the lower-level coefficients as auxiliary variables subject to three upper bounds. Taking each auxiliary coefficient to be the minimum in equation (8) is equivalent for feasibility and makes the finite certificate a single vector of principal weights.

For the theorem we take

$$k = 18, \qquad \gamma = \frac{901}{1000}, \qquad \lambda = 2^\gamma. \tag{12}$$

There are $3^{17} = 129{,}140{,}163$ principal rows. The exponent $901/1000$ leaves the reserve $1/1000$ needed for the unit-coefficient $9/10$ statement. It is a certified feasible value, not a claimed optimum of the level-18 program.

The level and exponent were chosen as a continuation of the preceding level-15 certificate, not by an exhaustive optimization theorem. Moving from level 15 to level 18 multiplies the principal state space by 27. The accepted exact run certified $901/1000$; no conclusion is drawn here about levels 16 or 17, or about infeasibility at any larger exponent.

TABLE 1. Exact level-18 LP certificate.

Quantity	Exact value
Principal weights / checked rows	129,140,163
Rows of each type	43,046,721
Failed rows	0
Minimum / maximum weight	1,048,576 / 1,859,404,226
Minimum row margin	4,147,742,755
Index / residue attaining it	18,958,255 / 56,874,767
Payload size	516,560,652 bytes
Payload SHA-256	`fc96181c552398be...ca7c34c`

3.1. **Integer formulation.** Put $Q = 2^{28} = 268{,}435{,}456$ and

$$A = 76{,}981{,}049, \qquad B_1 = 207{,}142{,}911, \qquad B_3 = 386{,}810{,}365. \tag{13}$$

The dyadic ratios are certified lower bounds:

$$\frac{A}{Q} \le \lambda^{-2}, \qquad \frac{B_1}{Q} \le \lambda^{\alpha-2}, \qquad \frac{B_3}{Q} \le \lambda^{\alpha-1}. \tag{14}$$

They are proved without floating point by the integer comparisons

$$A^{1000}2^{1802} \le Q^{1000}, \tag{15}$$

$$B_1^{1000}2^{1802} \le Q^{1000}3^{901}, \tag{16}$$

$$B_3^{1000}2^{901} \le Q^{1000}3^{901}. \tag{17}$$

Index principal residues by $i = 0, \ldots, 3^{17} - 1$, with $m_i = 3i + 2$. Let w_i be a positive integer weight, let $f(i)$ represent $4m_i \pmod{3^{18}}$, and let $u_1(i), u_3(i)$ represent the lower-level auxiliary residues. With $N_- = 3^{16}$, write

$$\underline{w}_u = \min\left\{w_u, w_{u+N_-}, w_{u+2N_-}\right\}.$$

The exact rows are

$$i \equiv 0 \pmod 3: \quad Qw_i \le Aw_{f(i)} + B_1\underline{w}_{u_1(i)}, \tag{18}$$

$$i \equiv 1 \pmod 3: \quad Qw_i \le Aw_{f(i)}, \tag{19}$$

$$i \equiv 2 \pmod 3: \quad Qw_i \le Aw_{f(i)} + B_3\underline{w}_{u_3(i)}. \tag{20}$$

Lemma 3.1 (Exact level-18 LP certificate)**.** *There is a vector $w \in \mathbb{N}^{3^{17}}$, with every $w_i > 0$, satisfying equations (15) to (20). Consequently $c_{m_i} = w_i$ is feasible for $L_{18}^{NT}(2^{901/1000})$.*

Proof. The binary vector bound by the hash in table 1 is checked for length and positivity. The checker then proves the three coefficient directions through equations (15) to (17), computes the residue maps and three-lift minima, and evaluates every row of equations (18) to (20). All checks pass and the least right-minus-left margin is positive. Since all weights are nonnegative, the coefficient bounds equation (14) imply the real LP rows term by term. □

The vector was found by integer normalized power iteration. Search is not a proof input: the accepted object is the explicit vector together with the small exact checker. Separate Python and Boost/C++ implementations agree on all row counts, the least margin, and the worst row.

This distinction between search and verification is important at this scale. The earlier computational literature reports numerical feasible values; the present theorem instead exposes an integer witness, exact coefficient directions, and complete row-by-row replay. The comparison concerns the certification format, not the correctness of those earlier numerical conclusions.

4. Adaptive advanced-term elimination

lemma 3.1 predicts the desired exponential lower bound, but the advanced D3 term prevents us from inserting it into an ordinary induction. This section converts each one-step source inequality into a finite *retarded* expression. The construction must serve two different purposes: its branches must be generous enough for coefficient transport, yet at the time of evaluation it must contain a branch that realizes the actual functional minimum. Keeping these roles separate is the main analytic point.

4.1. **Expression trees.** Let $I = \mathcal{P}_{18}$. Expressions are built from shifted leaves (i, β), addition, and minimum. For functions $F_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, evaluate a leaf by

$$(i, \beta)[F](y) = F_i(y + \beta),$$

and interpret addition and minimum literally. For positive weights c_i, define

$$\mathcal{C}_\lambda((i, \beta); c) = c_i \lambda^\beta,$$

again extending recursively through the same two operations. Consequently

$$E[(\Delta c_i \lambda^{\cdot})_i](y) = \Delta \lambda^y \mathcal{C}_\lambda(E; c). \tag{21}$$

We say that an expression has shift bounds (μ, ν) when every leaf shift β satisfies

$$-\nu \leq \beta \leq -\mu < 0. \tag{22}$$

After substituting the three-lift minimum equation (4), each source row equations (5) to (7) is an expression tree. A negative leaf is terminal. A nonnegative leaf is expanded by the source row for its residue. The issue is to prove that this recursive expansion cannot continue forever.

4.2. **Exact shift codes and the adaptive policy.** Every expansion label consists of a principal index and a code $(p, q) \in \mathbb{N}^2$ representing the exact shift

$$s(p, q) = p\alpha - q. \tag{23}$$

The root has code $(0, 0)$. The three edge types are

Edge	Code update	Shift increment
principal	$(p, q) \mapsto (p, q + 2)$	-2
D1 auxiliary	$(p, q) \mapsto (p + 1, q + 2)$	$\alpha - 2$
D3 auxiliary	$(p, q) \mapsto (p + 1, q + 1)$	$\alpha - 1$

Only a node with nonnegative shift is expanded. Irrationality of α ensures that a nonroot shift code is never zero.

Deletion is history-sensitive rather than a property of the residue alone. A fresh auxiliary lift v is *dominated* when $s(v) \geq 0$ and an earlier node u on its root path has

$$i(u) = i(v), \qquad s(u) < s(v). \tag{24}$$

An ordinary survivor is a lift that is not dominated. The expansion retains every survivor. If all three auxiliary lifts are dominated, it must still retain one lift so that the auxiliary minimum is nonempty. At level 18 this fallback is selected by the potential of the successor state.

The resulting child relation is finitely branching. A path is *legal* when every edge is retained by this policy and every node that is expanded has nonnegative shift. If expansion failed to terminate, Kőnig's lemma would give an infinite legal path. The purpose of the finite potential below is to exclude exactly such a path.

4.3. **Motivation for the adaptive fallback.** At level 15, retaining lift zero whenever all three auxiliary lifts were dominated produced a bounded termination potential. Applying the same fixed-lift work-list at level 18 produced continuing upward waves rather than a bounded fixed point; the other globally fixed fallback pairs tested in the computation behaved similarly. These diagnostics motivated the adaptive search, but they are not used as negative theorem evidence: we do not claim that every conceivable proof for a fixed fallback is impossible.

The successful rule is local and deterministic. In a D1 or D3 row, inspect the three successor values of P, choose a least one, and break ties in the order $0, 1, 2$. This selector is consulted only in the all-dominated case. When an ordinary survivor exists, every survivor is retained independently of the selector. That separation will later ensure that the certificate-driven fallback cannot displace the lift which realizes the actual functional minimum.

4.4. **The adaptive potential.** Let $F(i)$ be the principal successor and let $D_1(i,\ell), D_3(i,\ell)$ be the three auxiliary successors. For any candidate potential P, define

$$\ell_j^P(i) = \min\left(\operatorname*{arg\,min}_{0\le\ell<3} P(D_j(i,\ell))\right) \qquad (j \in \{1,3\}),$$

so ties are broken in the order $0, 1, 2$.

Lemma 4.1 (Adaptive forced-transition potential)**.** *There is a function*

$$P : \left\{0,\ldots,3^{17}-1\right\} \longrightarrow \{0,\ldots,34\}$$

such that

$$P(F(i)) \le P(i) + 6 \qquad \textit{for every } i, \tag{25}$$

$$P(D_1(i,\ell_1^P(i))) \le P(i) + 1 \qquad \textit{when } i \equiv 0 \pmod 3, \tag{26}$$

$$P(D_3(i,\ell_3^P(i))) + 2 \le P(i) \qquad \textit{when } i \equiv 2 \pmod 3. \tag{27}$$

Proof. The unsigned-byte vector is checked at all indices. There are 129,140,163 principal inequalities and 43,046,721 of each auxiliary kind, for 215,233,605 checks. Every entry lies between 0 and 34 and every inequality passes. The payload SHA-256 is

`778584ce7635639ab2a0c7de766f81e8f551b0a52f96ca51eb2bb128c3248f52`.

Separate exact Python and C++ verifiers reproduce the result. The C++ verifier also changes one potential byte and rejects the mutation with exactly one failed inequality. ☐

The selector is genuinely adaptive. In each of the D1 and D3 families its lift counts are

$$42{,}561{,}920, \qquad 484{,}085, \qquad 716 \tag{28}$$

for lifts 0, 1, and 2. Thus nonzero choices are rare but genuinely required by this particular certified potential.

For example, in the D3 row with principal index 83 (residue 251), the successor potentials are $(10, 9, 9)$. Least-index tie breaking selects lift 1 and the checked inequality is

$$P(D_3(83, 1)) + 2 = 9 + 2 = 11 = P(83).$$

The third lift is genuinely needed as well: at D3 index 78,929, the successor potentials are $(7, 7, 6)$, so lift 2 is selected and

$$P(D_3(78{,}929, 2)) + 2 = 6 + 2 = P(78{,}929).$$

These rows make the distinction from a globally fixed lift visible without requiring inspection of the full payload.

For a labelled node v, define

$$H(v) = 3s(v) + P(i(v)), \tag{29}$$

and put

$$\delta = 5 - 3\alpha > 0. \tag{30}$$

The positivity is elementary and exact:

$$3\alpha < 5 \quad\iff\quad 3^3 < 2^5 \quad\iff\quad 27 < 32.$$

The three potential bounds were chosen to match the exact shift increments. Along a principal edge,

$$\Delta H \leq 3(-2) + 6 = 0.$$

Along a selected D1 auxiliary edge,

$$\Delta H \leq 3(\alpha - 2) + 1 = 3\alpha - 5 = -\delta,$$

and along a selected D3 auxiliary edge,

$$\Delta H \leq 3(\alpha - 1) - 2 = 3\alpha - 5 = -\delta.$$

Thus principal motion does not increase H, while every selected fallback auxiliary edge consumes the same positive amount of adjusted-shift budget.

Lemma 4.2 (Syndetic rotation obstruction)**.** *Let $S \subseteq \mathbb{N}$ have bounded forward gaps, let $\kappa : S \to C$ take values in a finite set, and write $s_n = n\alpha - q_n \geq 0$, with $q_n \in \mathbb{N}$. It is impossible that*

$$n < n', \quad \kappa(n) = \kappa(n') \quad\implies\quad s_{n'} \leq s_n \tag{31}$$

for all $n, n' \in S$.

Proof. The fractional part of s_n is $\{n\alpha\}$. The antitone condition bounds the integer parts of s_n, separately for each color, by their value at the first occurrence of that color. Refine the color by this bounded integer part; only finitely many refined colors occur.

Suppose every interval $[m, m+L]$ meets S. Choose a short open interval $J \subset \mathbb{R}/\mathbb{Z}$ avoiding the finitely many cut points $-d\alpha \pmod 1$, $0 \leq d \leq L$. Translation by each $d\alpha$ is then order-preserving on J, with no wraparound. Divide J into many ordered subintervals. Density of every tail of the irrational rotation lets us choose chronological base indices, separated by more than L, whose fractional parts visit these subintervals from left to right. For each base choose a point of S at an offset $d \in \{0, \dots, L\}$. Pigeonholing the offsets leaves an arbitrarily long subsequence with one common d; its fractional parts remain strictly increasing.

A subsequence longer than the number of refined colors contains two terms with the same original color and integer part but increasing fractional parts. The corresponding s_n increase, contradicting equation (31). □

Theorem 4.3 (Adaptive level-18 elimination terminates)**.** *From every level-18 principal root, the adaptive expansion is finite. Every terminal node in the resulting full normal form has negative shift.*

Proof. Suppose there is an infinite legal path. Compress each maximal run of principal edges, retaining one endpoint for every auxiliary count n. At the n-th endpoint the exact shift is

$$s_n = n\alpha - q_n \geq 0$$

for an integer q_n, because an infinite legal path continues only through expanded nodes. Color the endpoint by its principal residue.

Call an endpoint a record if it is the root or if its auxiliary edge was an ordinary survivor. Two chronological records of the same color have nonincreasing shifts; otherwise the earlier equal-residue label would dominate the later one.

Every nonrecord endpoint must have arisen from the certificate-selected fallback, because ordinary survivors are records. Between auxiliary endpoints, principal edges do not increase H, and at each nonrecord endpoint H loses at least δ.

For each residue color, record shifts are bounded by the first record of that color. There are finitely many colors and $0 \leq P \leq 34$, so H has a uniform upper bound at every record. Away from records it only descends. On the other hand, every expanded node has nonnegative shift and nonnegative potential, hence $H \geq 0$. A record-free interval therefore cannot contain more than a fixed number of auxiliary endpoints: otherwise its repeated δ-loss would force $H < 0$. The record set is forward-syndetic.

Its shifts are classwise antitone, contradicting lemma 4.2. Hence no infinite legal path exists. The child relation is finitely branching, so well-founded recursion produces a finite full normal form from every root. Expansion stops exactly at negative shifts. ☐

Remark 4.4. The finite potential is needed only to make all-dominated fallback runs spend adjusted-shift budget. It does not decide which branch will be used in the functional inequality at a given time. That branch is chosen separately, from the actual envelope values, in the next subsection.

4.5. **Coefficient transport and time-local functional choice.** Fix one finite full normal form $\mathcal{T}_i$ for each principal root. Finiteness of the index set and all trees gives common constants

$$\mu > 0, \qquad \nu \geq 2, \tag{32}$$

such that every terminal shift β lies in

$$-\nu \leq \beta \leq -\mu < 0. \tag{33}$$

The full normal form should be viewed as a finite menu of coefficient-valid branches with common retarded bounds. It need not itself be the functional witness at every time.

The coefficient comparison is unaffected by adaptive selection. At an auxiliary node, the retained family consists of all survivors, or of the certificate-selected lift when there are no survivors. It is therefore a nonempty subset of the three source lifts. The minimum over a nonempty subset is at least the minimum over all three.

For completeness, coefficient transport is an induction over the finite normal form. LP feasibility bounds the coefficient of each expanded leaf by the coefficient value of its one-level source expression. The nonempty-subfamily observation preserves this bound at every auxiliary minimum, including an all-dominated node where only the selected fallback is retained. The induction hypotheses then replace retained advanced terminals by their normalized children. At the root this gives

$$c_i \leq \mathcal{C}_\lambda(\mathcal{T}_i; c). \tag{34}$$

The fixed full tree is not asserted to be a functional witness at every time. At an auxiliary minimum, the lift attaining the minimum of the actual functions ϕ_{18}^m may depend on y. We therefore prune the full tree anew at the current pair (i, y).

Lemma 4.5 (Dominated lifts are not critical minima)**.** *Suppose a critical source subtree has value*

$$p + \min_{v\in\mathcal{A}} F_{i(v)}(y + s(v)), \qquad p > 0,$$

and this value is at most every earlier path value $F_{i(u)}(y + s(u))$. A fresh lift dominated by an earlier u cannot attain the displayed minimum.

Proof. If a dominated lift v attained the minimum, positivity of p would give

$$F_{i(v)}(y + s(v)) < p + \min_{z\in\mathcal{A}} F_{i(z)}(y + s(z)) \leq F_{i(u)}(y + s(u)).$$

But domination gives $i(u) = i(v)$ and $s(u) < s(v)$, so monotonicity of the same envelope gives the reverse weak inequality. ☐

The upper-bound hypothesis in the lemma is maintained by a *critical-history invariant.* At the root the current source expression is bounded by its left-hand envelope. When we descend through the principal summand or an auxiliary child attaining the minimum, nonnegativity makes the chosen child no larger than its parent. Hence it remains bounded by every earlier label on the critical path.

It follows from lemma 4.5 that an actual minimizing lift is never dominated. It is therefore retained as an ordinary survivor, independently of which fallback the potential would select if all lifts were dominated. Following actual minima and discarding the other alternatives produces a *critical pruning*. Three facts are then immediate:

(i) its functional value is at most the left-hand envelope;
(ii) removing alternatives from a minimum can only increase $\mathcal{C}_\lambda$;
(iii) pruning preserves the common shift bounds equation (33).

The first fact is functional, the second is coefficient-theoretic, and the third supplies the common delay needed for induction. Their separation is why the adaptive fallback can be used safely.

Proposition 4.6 (Time-local choice witness)**.** *Let*

$$F_i(y) = \phi_{18}^{m_i}(y), \qquad c_i = w_i, qquad\lambda = 2^{901/1000}.$$

There are common $\mu > 0$ and $\nu \geq 2$ such that for every i and every $y \geq \nu$, one can choose an expression $E_{i,y}$ satisfying

$$\text{every leaf shift lies in } [-\nu, -\mu], \tag{35}$$

$$c_i \leq \mathcal{C}_\lambda(E_{i,y}; c), \tag{36}$$

$$E_{i,y}[F](y) \leq F_i(y). \tag{37}$$

Proof. Use the finite normal forms from theorem 4.3. Source feasibility and the nonempty-subfamily comparison give equation (34). At (i, y), follow actual minimizing lifts through the source expressions. The critical-history invariant and lemma 4.5 ensure that each selected lift occurs in the full adaptive normal form. Thus the critical pruning gives equation (37). Pruning raises the coefficient value relative to equation (34), giving equation (36), and preserves the terminal leaves, giving equation (35). □

Remark 4.7 (Reusable adaptive-elimination interface)**.** The new certificate-dependent input to the preceding construction consists of two finite objects: positive feasible coefficients and a bounded potential whose selected fallback edges decrease the adjusted shift by a fixed amount. History domination then supplies classwise antitonicity, irrational rotation forces termination, and critical pruning recovers a time-local minimizing branch without weakening coefficient transport. Thus the reusable mechanism is

$$\boxed{\begin{array}{c}\text{finite potential} + \text{strict fallback loss}\\ + \text{ history domination}\\ \Downarrow\\ \text{retarded time-local choice witnesses}\end{array}}$$

Higher levels may replace the two finite certificates while retaining this analytic interface.

The expression $E_{i,y}$ may depend on both i and y. No globally fixed pruned tree is asserted to realize every functional minimum. What is uniform is the finite menu, the coefficient lower bound, and the interval of negative terminal shifts. The next lemma is designed precisely for this choice-valued interface.

5. Choice-valued retarded growth

Lemma 5.1 (Choice-valued retarded induction)**.** *Let I be finite. Suppose $F_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ are nondecreasing, $c_i > 0$, and $\lambda > 1$. Suppose there are $\mu > 0$ and $\nu \geq 0$ such that for every i and $y \geq \nu$ one can choose an expression satisfying equations (35) to (37). If $F_i(0) \geq b > 0$ and $c_i \leq C_{\max}$, then*

$$F_i(y) \geq \frac{b\lambda^{-\nu}}{C_{\max}} c_i \lambda^y \qquad (i \in I,\ y \geq 0). \tag{38}$$

Proof. Set $\Delta = b\lambda^{-\nu}/C_{\max}$. For $0 \le y \le \nu$, monotonicity gives

$$F_i(y) \ge b \ge \Delta c_i \lambda^y.$$

Induct over intervals of length μ. Suppose the estimate is known through time $\nu + r\mu$, and take a new $y \le \nu + (r+1)\mu$. Choose the expression $E_{i,y}$ supplied at that specific time. Every leaf time $y + \beta$ lies in the previously controlled interval: the upper shift bound gives $y+\beta \le \nu+r\mu$, while the lower bound and $y \ge \nu$ keep the invoked times nonnegative after the initial interval is handled. Monotonicity of addition and minimum, followed by equation (21), yields

$$E_{i,y}[F](y) \ge \Delta\lambda^y \mathcal{C}_\lambda(E_{i,y}; c) \ge \Delta c_i \lambda^y.$$

Now use equation (37). $\square$

Taking $b = 1$ and $C_{\max} = \sum_i w_i$ gives the analytic consequence of the two certificates.

Theorem 5.2 (Level-18 envelope growth). *There is $\Delta > 0$, independent of i, such that*

$$\phi_{18}^{m_i}(y) \ge \Delta w_i \left(2^{901/1000}\right)^y \tag{39}$$

for all i and $y \ge 0$.

Proof. Apply proposition 4.6 and lemma 5.1. One may take $\Delta = \lambda^{-\nu}/\sum_i w_i > 0$. $\square$

No numerical value of ν enters the exponent. Its existence changes only the positive prefactor Δ. This explains why a finite but unmaterialized family of normal forms suffices for the asymptotic theorem, while also explaining why the proof does not presently yield a practical threshold.

6. From envelopes to every fixed target

The envelopes exclude periodic targets and use principal residues congruent to $2 \pmod 3$, whereas theorem 1.1 permits periodic targets and both unit residue classes. The following elementary reduction closes both gaps without assuming that any orbit reaches 1.

Lemma 6.1 (Nonperiodic source reduction). *For every $a > 0$ with $3 \nmid a$, there is a positive nonperiodic integer $b \equiv 2 \pmod 3$ and an $r \ge 0$ such that $T^r(b) = a$.*

Proof. If $a \equiv 2 \pmod 3$ is nonperiodic, take $b = a$. If $a \equiv 1 \pmod 3$ and $2a$ is nonperiodic, take $b = 2a$.

If periodic points x, y have a common iterate $T^r(x) = T^r(y)$, choose a later time divisible by both periods and apply the remaining iterates to deduce $x = y$. Now suppose $a \equiv 2 \pmod 3$ has period $p > 0$, and take

$$b = 2^{2p}a.$$

Then $b \equiv a \equiv 2 \pmod 3$ and $T^{2p}(b) = a = T^{2p}(a)$. If b were periodic, the common-iterate observation would give $b = a$, contradicting $b > a$.

Finally suppose $a \equiv 1 \pmod 3$ and $2a$ is periodic. Then a is periodic as well. For a period $p > 0$, take

$$b = 2^{2p+1}a.$$

Now $b \equiv 2 \pmod 3$, and both b and $2a$ map to a after $2p + 1$ iterates. If b were periodic, the same common-image argument would give $b = 2a$, again impossible. In every case the chosen power-of-two multiple reaches a. $\square$

Proof of theorem 1.2. Choose b by lemma 6.1 and let m_i be its residue modulo 3^{18}. Since b is one of the targets over which $\phi_{18}^{m_i}$ is minimized, for $y \ge 0$,

$$\phi_{18}^{m_i}(y) \le \pi_b^*(2^y b) \le \pi_b(2^y b) \le \pi_a(2^y b). \tag{40}$$

TABLE 2. Finite proof obligations and exact replay.

Certificate	Inequalities	Failures	Verifiers
Level-18 LP weights	129,140,163	0	Lean, Python, C++
Adaptive potential	215,233,605	0	Lean, Python, C++
Total	344,373,768	0	exact arithmetic

The last inequality is composition of reachability: every predecessor reaching b subsequently reaches a. For $x \geq b$, set $y = \log_2(x/b)$, so $2^y b = x$. Combining equations (39) and (40) gives

$$\pi_a(x) \geq \frac{\Delta w_i}{b^{901/1000}} x^{901/1000}.$$

Thus $C_a = \Delta w_i / b^{901/1000} > 0$ is admissible. □

Proof of theorem 1.1. Eventually

$$\pi_a(x) \geq C_a x^{901/1000} = x^{9/10}(C_a x^{1/1000}),$$

and the parenthesized factor is at least one once $x \geq C_a^{-1000}$. Together with the threshold in theorem 1.2, this proves the claim. □

For $a = 1$, this says that at least $x^{0.90}$ positive integers below x eventually reach the cycle $1 \leftrightarrow 2$. It says nothing about the remaining integers and does not imply global convergence.

7. Exact computation and formal verification

7.1. **Finite arithmetic.** The two native payloads deliberately have simple semantics.

(1) The LP payload is an array of 3^{17} little-endian unsigned 32-bit weights. A verifier checks its length, positivity, the three integer power comparisons equations (15) to (17), every residue map, every three-lift minimum, and all rows equations (18) to (20).
(2) The adaptive-potential payload is an array of 3^{17} unsigned bytes. A verifier checks $P(i) \leq 34$, computes the least-potential lift with the stated tie break, and evaluates every inequality equations (25) to (27).

Neither replay uses floating-point arithmetic. The decisive totals are:
The complete payload hashes are

LP weights.
`fc96181c552398be8bbc4c0a47482a53c4ff847059bf7fd4b9be11837ca7c34c`

Adaptive potential.
`778584ce7635639ab2a0c7de766f81e8f551b0a52f96ca51eb2bb128c3248f52`

The potential verifier's deliberate mutation test is useful evidence that the implementation evaluates the inequalities rather than only checking metadata or a file hash. The Python and C++ programs are separate implementations of the finite semantics. Neither finite replay proves the analytic implication; that implication is formalized deductively in Lean.

The exact verifiers and Lean checks have complementary roles. The standalone programs make the elementary finite claims independently inspectable and can be reimplemented without Lean. Lean's native decisions connect those same payloads to the formal certificate predicates, while the kernel checks the deductive chain from those predicates to the public theorem. Agreement among the layers is therefore stronger than rerunning one implementation twice.

7.2. **Lean theorem chain.** The development uses Lean 4 [9] and Mathlib [8]. It formalizes the accelerated map, actual finite predecessor sets, envelopes, difference inequalities, both certificate interfaces, adaptive termination, coefficient transport, time-local critical pruning, retarded induction, target reduction, and exponent absorption. No theorem asserting the Collatz conjecture is used.

The central definitions and publication-facing statement are shown below. The namespaces are included because the broader repository also contains a standard-map reachability predicate with a different meaning.

```
namespace Erdos1135.Terras

def accelerated (n : ℕ) : ℕ :=
  if Even n then n / 2 else (3 * n + 1) / 2

end Erdos1135.Terras

namespace Erdos1135.KrasikovLagarias

def Reaches (source target : Nat) : Prop :=
  ∃ steps : Nat,
    Terras.accelerated^[steps] source = target

noncomputable def predecessorFinset
    (target ceiling : Nat) : Finset Nat := by
  classical
  exact (Finset.Icc 1 ceiling).filter fun source =>
    Reaches source target

noncomputable def predecessorCount
    (target ceiling : Nat) : Nat :=
  (predecessorFinset target ceiling).card

noncomputable def predecessorCountReal
    (target : Nat) (cutoff : Real) : Nat :=
  predecessorCount target ⌊cutoff⌋₊

theorem eventually_rpow_9_10_le_predecessorCountReal
    {target : Nat} (htarget : 0 < target)
    (hmod : target % 3 ≠ 0) :
    ∀ᶠ x in atTop,
      x ^ ((9 : Real) / 10) ≤
        (predecessorCountReal target x : Real)

end Erdos1135.KrasikovLagarias
```

Thus reachability permits zero steps. The fully qualified theorem is
`Erdos1135.KrasikovLagarias.K18PredecessorAllTargets.eventually_rpow_9_10_le_predecessorCountReal`.

The natural-cutoff form and the positive-constant 901/1000 form are also formalized.

The final module's local import closure contains 67 modules. The recorded `leanchecker` command checks every declaration in the final module against its compiled environment; it is not a separate source traversal of all 67 modules.

All three public declarations have the same printed axiom footprint:

`propext`, `Classical.choice`, `Quot.sound`, together with exactly two generated native-decision assertions:

`Erdos1135.KrasikovLagarias.k18AdaptiveForcedPotentialCertificate_check._native.native_decide.ax_1_1`

`Erdos1135.KrasikovLagarias.k18Gamma901EncodedCertificate_check._native.native_decide.ax_1_1`.

These are explicit compiler trust points. Their complete underlying integer computations are the ones replayed by the Python and C++ verifiers. The unrelated admitted declaration `CollatzConjecture.collatz_conjecture` is absent from the theorem's axiom closure.

7.3. **Reproducibility.** The formal source is frozen at commit

`0bb368e3b51b980ef5c3b33bb86316e3001c37ac`.

The successful environment used Lean `4.30.0-rc2`, commit `3dc1a088b6d2d8eafe25a7cd7ec7b58d731bd7cc`, and Mathlib commit `5450b53e5ddc75d46418fabb605edbf36bd0beb6`. The complete preservation manifest has SHA-256

`6120dd80c95b24b30f440b4394f736db0945271082ed69ced5eddb9f0e80b459`.

It binds the Git bundle, both raw payloads, the Python and C++ verifiers, their reports, the Lean validation reports, and the compact review surface.

The source bundle identifies the same commit as the manuscript, and every one of the 20 entries in the preservation manifest passed a fresh checksum verification. The raw files are omitted from the small paper bundle only because of size; the hashes in this paper and the manifest fix their exact contents.

The first-party project page is

https://www.proofatlas.ai/collatz-predecessor-090/.

Because the two raw payloads total 645,700,815 bytes, the preservation layout permits a compact source/reviewer archive to be distributed separately from the optional full replay payload. The checksum manifest binds the two tiers as one reproducible release.

8. Context, limitations, and further questions

This theorem strengthens a quantitative record within an established method for inverse trees. It is logically different from forward-orbit and almost-everywhere small-value results, such as Tao's logarithmic-density theorem [11]. The standard conjecture on predecessor density asks for $\pi_a(x) \geq c_a x$, so exponent 0.90 remains a waypoint rather than the expected endpoint.

The value 901/1000 is feasible, not claimed optimal at level 18. The computation proves no infeasibility statement for larger exponents. The original program asks whether optimal bases tend to 2, which would give $x^{1-\varepsilon}$ predecessor bounds for every $\varepsilon > 0$. The present theorem does not resolve that limiting question.

The adaptive potential is level-specific data, although its Lean consumer is generic in the level and in the selected lifts. Consequently a higher-level certificate can reuse the termination and time-local proof without reverting to a globally fixed fallback. A structural family of potentials, or an analytic construction valid uniformly in k, would be a more substantial advance than another isolated finite level.

Finally, the constants and thresholds are not extracted. In particular, the common retarded bound ν comes from finite-normal-form existence and is not evaluated. The theorem is effective in principle at fixed level but not effective as proved here.

Three extensions are now cleanly separated.

(1) At fixed level 18 one can optimize the feasible point or certify a larger exponent. This changes the finite LP input but not the analytic theorem.
(2) At higher levels one can search simultaneously for a feasible weight vector and an adaptive potential. The generic Lean consumer already permits state-dependent selected lifts, so the new proof mechanism is reusable.

(3) A uniform construction of potentials, or a direct structural termination theorem independent of a finite table, could address the limiting question $\lambda_k \to 2$. That would be qualitatively stronger than merely adding another level.

Computing the finite normal forms themselves would be a different practical improvement: it would make ν, the prefactor, and ultimately an explicit target threshold available, without changing the exponent.

Appendix A. Certificate schema and worked rows

Let

$$N = 3^{17}, \qquad M = 3^{18}, \qquad N_- = 3^{16},$$

and index $0 \le i < N$ by $m = 3i + 2$. Define

$$\begin{aligned} f(i) &= \frac{(4m \bmod M) - 2}{3}, \\ u_1(i) &= \frac{(((4m-2)/3 \bmod 3^{17}) - 2)}{3}, \\ u_3(i) &= \frac{(((2m-1)/3 \bmod 3^{17}) - 2)}{3}. \end{aligned}$$

The auxiliary successor maps used by the adaptive checker are therefore

$$D_j(i,\ell) = u_j(i) + \ell N_- \qquad (j \in \{1,3\},\ 0 \le \ell < 3).$$

The row class is determined by $i \bmod 3$. Thus this appendix specifies the indices needed to reimplement both the LP row predicate and the adaptive least-potential selection without reading the certificate generator.

With the constants in equation (13), the first row of each class is

$$\begin{aligned} L1\ (i=0, m=2):\quad 2{,}134{,}179{,}986{,}800{,}640 &\le 76{,}981{,}049 \cdot 6{,}357{,}317 \\ &\quad + 207{,}142{,}911 \cdot 7{,}945{,}915 \\ &= 2{,}135{,}332{,}895{,}144{,}098, \\ L2\ (i=1, m=5):\quad 1{,}036{,}938{,}517{,}676{,}032 &\le 76{,}981{,}049 \cdot 13{,}483{,}121 \\ &= 1{,}037{,}944{,}798{,}373{,}929, \\ L3\ (i=2, m=8):\quad 1{,}706{,}529{,}287{,}831{,}552 &\le 76{,}981{,}049 \cdot 2{,}896{,}800 \\ &\quad + 386{,}810{,}365 \cdot 3{,}838{,}338 \\ &= 1{,}707{,}707{,}625{,}516{,}570. \end{aligned}$$

For L1, the three auxiliary indices are $(0, 43{,}046{,}721, 86{,}093{,}442)$, and the minimum weight occurs at the middle index. For L3 they are $(1, 43{,}046{,}722, 86{,}093{,}443)$, again with the middle weight minimal.

The first potential rows read

$$P(f(0)) = P(2) = 2 \le P(0) + 6 = 6, \qquad P(D_1(0,0)) = 0 \le 1,$$

and

$$P(D_3(2,0)) + 2 = P(1) + 2 = 2 \le P(2) = 2.$$

A genuinely adaptive example occurs already in the D3 row $i = 83$, residue 251. Its three successor potentials are $(10, 9, 9)$; least-index tie breaking selects lift 1, and

$$P(D_3(83,1)) + 2 = 9 + 2 = 11 = P(83).$$

Lift 2 is also used: at D3 index 78,929, the successor potentials are $(7, 7, 6)$ and $6 + 2 = P(78{,}929) = 8$.

Table 3. Proof chain, Lean modules, and trust mode.

Mathematical surface	Lean module	Check
Predecessor counts and envelopes	`PredecessorCount.lean`, `Envelope.lean`	deductive
Source difference system	`EliminationSourceSystem.lean`, `K18SourceBridge.lean`	deductive
Level-18 feasible point	`Generated/K18Gamma901.lean`, `K18CoefficientBridge.lean`	native + replay
Adaptive potential and policy	`AdaptiveForcedPotentialCertificate.lean`, `Generated/K18AdaptiveForcedPotential.lean`	native + replay
No infinite adaptive path	`AdaptiveAuxiliaryTermination.lean`, `AdaptivePathExtraction.lean`	deductive
Coefficient transport and critical pruning	`AdaptiveNormalizedCoefficient.lean`, `AdaptiveCriticalSkeleton.lean`	deductive
Choice-valued retarded growth	`AdaptiveCriticalChoice.lean`, `K18CriticalChoice.lean`	deductive
Fixed-target transfer and final theorem	`K18PredecessorBridge.lean`, `K18PredecessorAllTargets.lean`	deductive

Appendix B. Formal correspondence

The constant-factor declaration is

`eventually_const_mul_rpow_901_1000_le_predecessorCountReal`,

and the natural-cutoff form of theorem 1.1 is

`eventually_natCast_rpow_9_10_le_predecessorCount`.

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